SNUNet3+: A Full-Scale Connected Siamese Network and a Dataset for Cultivated Land Change Detection in High-Resolution Remote-Sensing Images

Lizhi Miao, Xinting Li, Xinxin Zhou, Ling Yao, Yamei Deng, Tian Hang, Yuchao Zhou, and Haozhou Yang

***Abstract*— The decline of cultivated land significantly threatens the food supply. In recent years, remote sensing (RS) change detection emerged as a valuable tool for monitoring nonagriculturalization. However, existing deep learning (DL) methods for change detection suffer from the problem of inadequate utilization of feature information during image feature extraction, leading to noisy or inaccurate change maps. To address these challenges, we propose a Siamese network based on full-scale connected UNet (SNUNet3+), whose encoder extracts features from the original images at various levels. Then, it merges all fine-grained spatial information and coarse-grained semantic information into the decoder through full-scale skip connections to obtain the feature maps with complete information. The spatial and channel squeeze and excitation (scSE) attention mechanism is embedded within the decoder subnetwork to enhance the discriminative power of the features. In addition, a deep supervision module is introduced to improve the feature learning capability of the hidden layers and the quality of features. Moreover, accurate cultivated land change detection remains challenging because of the lack of fine-grained detection datasets. To enable the proposed method to achieve cultivated land change detection, we produced a cultivated land change detection dataset containing 5170 pairs of 256 × 256 bitemporal images with a spatial resolution of 1 m. The effectiveness and performance of the proposed method are evaluated on three high-resolution RS change detection datasets. Extensive experimental results show that the proposed method outperforms other state-of-the-art methods, achieving the highest *F*1-score of 72.90%, 90.36%, and 96.64% on the CLCD dataset, LEVIR-CD dataset, and PX-CLCD dataset, respectively.**

***Index Terms*— Change detection, deep learning (DL), nonagriculturalization of cultivated land, remote sensing (RS), UNet3+.**

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I. Introduction

With the rapid economic development and the acceleration of urbanization, the demand for land resources in cities has increased significantly, leading to a strain on land resources [1]. Cultivated land nonagriculturalization, such as nonagricultural construction, lake landscaping, and greening afforestation, have caused the decline of cultivated land in China [2]. In order to safeguard the "red line" of 120 million hectares and ensure food security, the Chinese government attaches great importance to the supervision and management of cultivated land [3]. Traditional methods for monitoring land resources rely heavily on manual on-site surveys, which are resource-intensive regarding human labor, materials, and finances. Developing intelligent methods for real-time updates in land surveys is urgently needed to assess the utilization of natural resources and monitor changes in land use and land cover (LULC) timely and accurate [4]. The advancements in remote sensing (RS) sensors have enabled the acquisition of high temporal resolution, high spatial resolution, and large-scale coverage of RS images [5]. Multitemporal high-resolution RS images provide rich spatial–temporal information and are extensively used in LULC change detections for a specific study area [6], [7]. However, the rapid and accurate detection of land use changes over large regions using high-resolution images remains a challenging task [8].

Traditional RS change detection methods can be mainly categorized into three types: 1) image algebra-based method; 2) image transformation; and 3) postclassification comparison. The image algebra-based method employs algebraic approaches to compare multitemporal images at a pixel level and classify pixels into "change" classes or "no-change" classes based on the thresholds for the difference value of pixels. This type directly compares pixel values, such as image difference [9], image regression [10], image ratio [11], and change vector analysis (CVA) [12]. Image transformation methods apply change analysis to vectors after dimensionality

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reduction of multitemporal images, including principal component analysis (PCA) [13] and multivariate alteration detection (MAD) [14]. The postclassification change method first classifies the pixels of the original image into different categories. Then, a difference map is generated by comparing the classification maps. For example, Peiman [15] applied PCA and band combination to preclassify the original images and conducted land use cover change detection in Pisa, Italy, by comparing the classification maps. Later, machine learning methods that learn change patterns from training samples were also used for RS change detection, such as support vector machines [16], decision trees [17], and random forests [18]. However, these methods often require the manual selection of features or thresholds for change detection, which can introduce subjectivity and errors, thereby impacting the accuracy and reliability of the change detection.

In recent years, due to the advancement in RS big data and artificial intelligence technology, the field of change detection has witnessed a shift from traditional machine learning-based methods to deep learning (DL)-based methods [19]. Convolutional neural network (CNN) has gained prominence because of their ability to extract high-dimensional information automatically, encompassing low-level spatial details and high-level semantic features from images. CNN and its variants, such as FCN [20], UNet [21], and ResNet [22], have achieved state-of-the-art performance in various visual tasks. Thanks to its excellent feature extraction capabilities, CNN is also used in many RS application scenarios. In the field of change detection, many researchers have leveraged CNNs to extract deep features from images and employ comparative analysis to obtain change maps, leading to excellent detection results [23], [24], [25]. Consequently, most recent studies in change detection have been founded on CNN-based methodologies.

DL-based frameworks for change detection can be categorized into single- and dual-branch structures based on their feature extraction strategies. The single-branch structure utilizes early fusion to merge bitemporal images into multispectral images. However, such a method faces challenges in extracting deep features from the original image, resulting in a change map that lacks regions with semantic information changes [19]. The dual-branch structure, a Siamese network, comprises two subnetworks with shared weights. Each branch independently learns higher level abstract feature representations of monotemporal images through multiple convolution and pooling layers. This robust representation capability enables the Siamese network to capture change information in complex RS images effectively, enhancing change detection accuracy. The Siamese network structure, designed for metric learning, facilitates the comparison of differences between two images [26]. For instance, Zagoruyko and Komodakis [27] incorporated the Siamese network with CNN to assess the similarity of two input images, achieving superior performance compared with previous DL models. Daudt et al. [28] proposed two Siamese networks, namely, FC-Siam-conc and FC-Siam-diff, employing fully convolutional shared weights. These networks utilized concatenation and differential feature fusion and were evaluated on two public datasets to validate the effectiveness of Siamese networks for RS change detection.

Many researchers have incorporated techniques to enhance feature extraction into DL change detection models and obtain better deep features. For example, the spatial or channel attention mechanisms [29] have been applied to capture more detailed and distinctive features. Moreover, the deep supervised networks [30] are added as auxiliary classifiers to the hidden layers of the models to improve the directness and transparency of the learning process. For instance, Zhang et al. [31] introduced a deep supervision module into the fully convolutional two-stream network to address potential gradient disappearance in the hidden layers during model training. They incorporated the convolutional block attention module (CBAM) to fuse multilayer depth features from the original image with the image difference features. Chen and Shi [32] proposed a spatial–temporal self-attentive mechanism for a CNN-based Siamese network. This module enables the computation of spatial–temporal dependencies between any two pixels of bitemporal images to generate more discriminative feature maps and provided a new and challenging dataset for building change detection. Chen et al. [33] developed a dual-attention fully convolutional Siamese network to obtain a more distinguishable feature representation and enhance the recognition performance of the model. They also proposed a weighted double-margin contrastive loss to address the sample imbalance issue in the detection process. Fang et al. [34] proposed a Siamese nested UNet (UNet++) structure for high-resolution image change detection, incorporating dense skip connections between the encoder and decoder. Compared with current state-of-the-art methods, this approach significantly improved evaluation metrics by preserving high-resolution features and fine-grained localization information while reducing uncertainty in edge pixels of change targets and minimizing false positives for small targets.

CNN captures local features from images through local receptive fields and weight sharing, then gradually extracts higher level semantic information. In RS change detection, CNN can effectively capture features, such as texture and shape, which is especially suitable for fine-grained change detection tasks, such as recognizing changes in roads and buildings, whereas, the transformer, as a DL architecture that relies on the parallel multihead attention mechanism, is primarily used to process sequential data, such as natural language text. Given that the transformer can capture complex relationships between pixels in RS images and enhance the original pixel-level features by modeling the spatial–temporal contextual information, many studies have utilized the transformer for RS change detection research in recent years. For example, Chen et al. [35] proposed a bitemporal image transformer (BIT) to efficiently and effectively model contexts within the spatial–temporal domain. They express bitemporal images as several tokens and use the transformer encoder to model context in a compact token-based spatial–temporal domain. The learned tokens enriched with global information are then fed back into the pixel space to refine the original features through the transformer decoder. Bandara and Patel [36] proposed a transformer-based Siamese network architecture

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for change detection from a pair of co-registered RS images. They found that the dependence on CNN for RS change detection is not essential and that the change detection task can be accomplished using only a transformer encoder paired with a lightweight multilayer perception (MLP) decoder. To leverage the advantage of CNNs in extracting local information, Li et al. [37] proposed a new CNN-transformer network with a multiscale framework to better exploit global–local information in optical RS images. Liu et al. [38] proposed a CNN-transformer network with multiscale global–local context aggregation for cultivated land change detection. However, since RS images typically have high resolution and large size, using transformers to process the entire image usually requires significant computational resources and is not particularly suitable for extracting some detailed features. Overall, both transformer and CNN have their respective advantages in RS image interpretation: transformers are suitable for capturing global information and establishing long-range dependency, which is more favorable for large-scale pattern recognition. On the other hand, CNNs excel in capturing local features and details and are suitable for fine-grained change detection tasks. In this article, we consider the detection of nonagriculturalization of cultivate land as fine-grained binary change detection, with the aim to accurately segment the changed objects on cultivated land. Although the ability of transformer to capture global features is better than CNN, however, the scale of each changed object in RS change detection is generally small. Therefore, it is more meaningful to use CNN to acquire local features than to acquire global features.

However, the existing CNN-based change detection methods often suffer from the inadequate utilization of extracted features. They have the limitation in effectively combining global and local features during continuous upsampling and downsampling, which results in the loss of spatial positioning information and object semantic information. This can lead to inaccurate results and pseudochanges. Previous models, such as Siamese UNet [28] and UNet++ [34], have shown promising performance in RS change detection tasks. However, they still cannot fully exploit all the information extracted by the network from lower layers to deeper layers from full scales. In contrast, UNet3+ [39] made improvements by redesigning the interconnection between the encoder and decoder and the intraconnection between decoders, enabling better utilization of fine-grained spatial information and coarse-grained semantic information. Inspired by these advancements, we propose a Siamese UNet3+ network for high-resolution image change detection in this article. The proposed network architecture uses a weight-shared feature extractor to extract features at different scales from bitemporal images. These features are then fused through a concatenation operation, and features of different scales are combined through full-scale skip connections. Furthermore, the spatial and channel squeeze and excitation (scSE) attention mechanism and the deep supervision module are incorporated during the decoding process to enhance the accuracy of the change detection results.

The performance of DL models heavily relies on the availability of large and accurately labeled datasets. Various change detection datasets have been developed, including Hi-UCD for semantic change detection [40], LEVIR-CD [32], WHU building change detection dataset [41], and S2Looking [42] for building change detection, and CLCD [38] for cropland change detection. However, the datasets specifically focused on cultivated land change detection are still limited. To address this issue, we have created a cultivated land change detection dataset called Peixian-cultivated land change detection (PX-CLCD) to enable our model to identify changes occurring on cultivated land effectively and accurately.

The main contributions of this research are summarized as follows.

1) A deeply supervised attention Siamese network, SNUNet3+, is proposed based on full-scale connected UNet for cultivated land change detection. The Siamese UNet3+ structure is used in the network backbone of SNUNet3+, and the scSE blocks are integrated as a joint attention mechanism in the decoder subnetwork. The deep supervision module enables the network training to focus on the full-scale feature information and generate more accurate and comprehensive change detection results.
2) A challenging cultivated land change dataset PX-CLCD was produced, which consists of 5170 pairs of 256×256 high spatial resolution 1-m RGB images and the corresponding binary change label maps. The dataset is available at: https://github.com/niuzhan/Peixian-Cultivated-land-Change-detection-dataset.

The remaining sections of this article are organized as follows. Section II introduces the study area and presents an overview of the proposed dataset, PX-CLCD. Section III will detail the general framework of the Siamese UNet3+ network and its internal structure. Section IV will introduce the datasets and methods used for the experiments, the parameters set for the model training, and the evaluation of the experimental results. Section V discusses the performance of scSE and deep supervision modules in SNUNet3+. Finally, we conclude this article in Section VI.

II. Materials

A. Study Area Overview

The study area for this article is Pei County, located in Jiangsu, China. It is situated in the northwest of Xuzhou City, with an area of about 1806 km^2, bounded by 34°28′–34°59′ N and 116°41′–117°09′ E. The specific geographical location is shown in Fig. 1. The county boasts a strategic location, a favorable ecological environment, and abundant resources, making it a significant grain production region. Given its typicality, cultivated land nonagriculturalization and any other changes occurring in the cultivated land within Pei County must be detected in time to protect food safety.

B. Data preprocessing

The high-resolution images used to create the dataset for this research are obtained from the Map World and captured by Gaofen-2 satellite in 2018 and 2021. The spatial resolution of the images ranges from 0.5 to 1 m, providing detailed and accurate information for the study area. Several preprocessing

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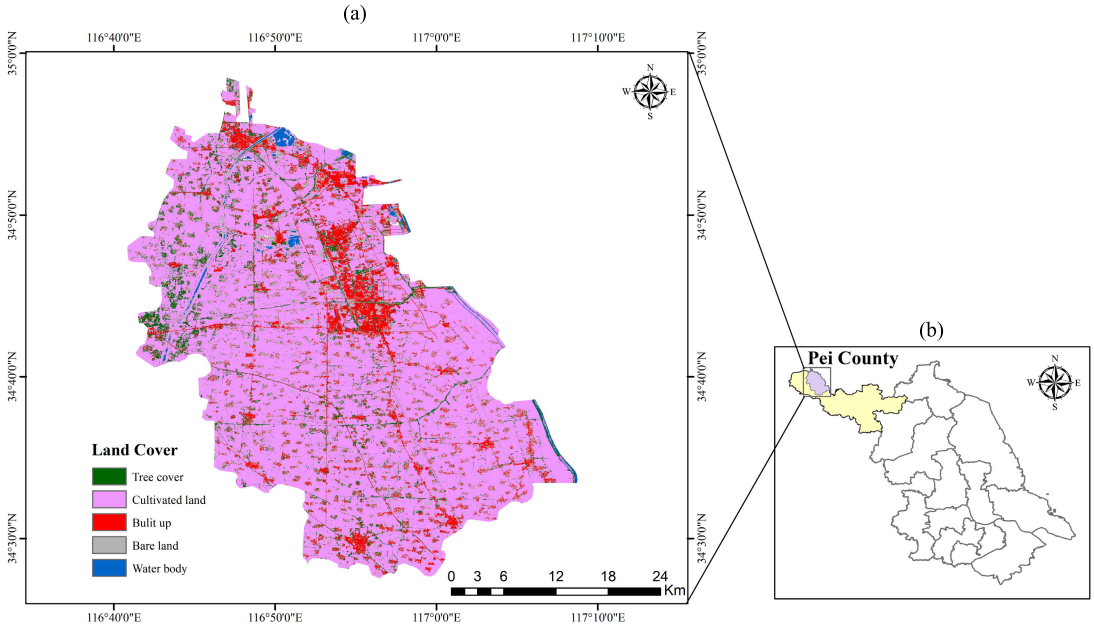


Fig. 1. Location and land cover of the study area. (a) Land cover map of Pei County in 2020. (European Space Agency Global Land Cover Product, 2020.) (b) Location of Pei County in Jiangsu Province.

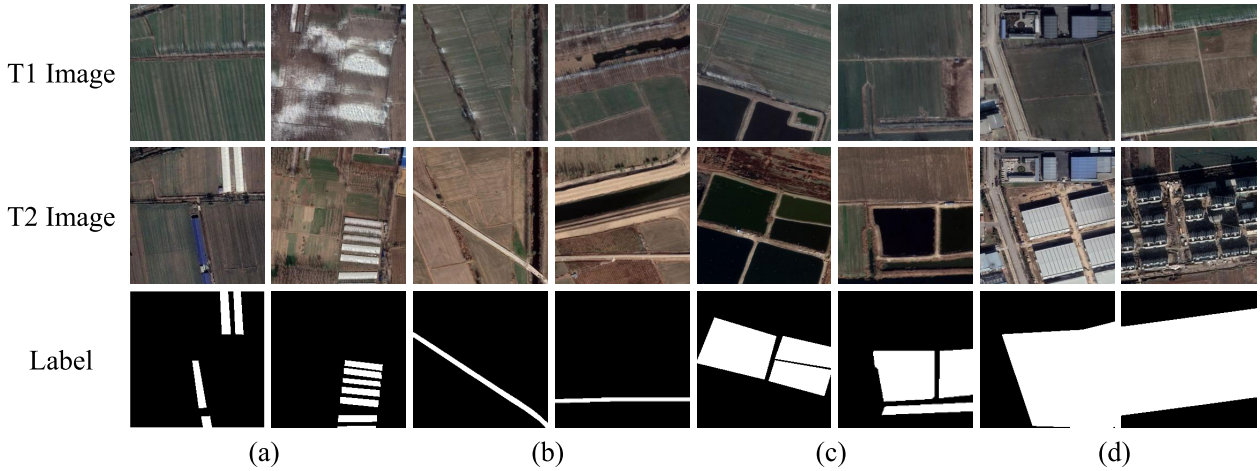


Fig. 2. Example samples (256 × 256) from PX-CLCD dataset. Each column represents a different change type in PX-CLCD. (a) Greenhouses, (b) roads, (c) lakes, and (d) houses.

operations were performed to prepare the original images for our research, including radiometric calibration, image registration, and image enhancement. First, the image was converted to satellite reflectance using the radiometric calibration tool in PIE-Basic software. Subsequently, the original bitemporal images were registered using the PIE-Basic image registration function. Finally, to facilitate the annotations of change areas, histogram equalization was applied to enhance the image, improving the visibility of the relevant features in the images.

C. Peixian Cultivated Land Change Detection (PX-CLCD) Dataset

The original dataset containing 5170 pairs of 1-m spatial resolution 256 × 256 bitemporal images, was divided into training, validation, and testing sets in a ratio of 6:2:2. The dataset was augmented by applying random flipping and rotation techniques to generate the final cultivated land change detection dataset PX-CLCD, respectively, representing 2018 and 2021. As shown in Fig. 2(a)–(d), the types of change in the PX-CLCD dataset mainly include: new constructions of greenhouses on cultivated land, new hardened roads on cultivated land, new constructions of artificial lakes on cultivated land, and new constructions of houses on cultivated land, respectively. We used the weights of SNUNet3+ trained on PX-CLCD to make predictions on a pair of bitemporal RS images and verified their effectiveness in identifying nonagriculturalization changes and the validity of the dataset.

After constructing the PX-CLCD dataset, we utilized a pair of high-resolution bitemporal images as the test area. The bitemporal images were captured by Gaofen-2 in Pei Country, China, which covers an area of about 1806 km^2, in 2018 and 2021, respectively. Cultivated land change detection is conducted to identify nonagricultural patches in the test

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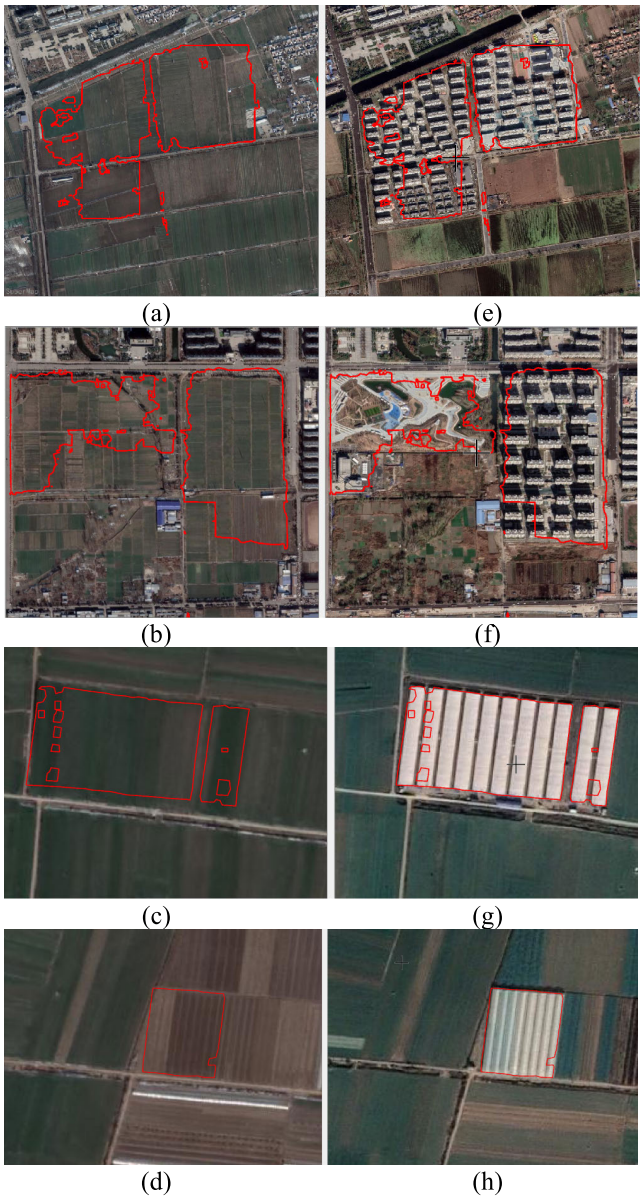


Fig. 3. Examples of the nonagricultural detection based on the PX-CLCD dataset. (a)–(d) are well-preserved cultivated lands and (e)–(h) are nonagricultural changes.

area using the model trained on the PX-CLCD dataset. Fig. 3 shows some identification results of the test area, where the red patches represent the "nonagriculturalized areas." Fig. 3(a)–(d) exhibits well-preserved cultivated lands from the previous period, whereas Fig. 3(e)–(h) depicts the artificial structures that have illegally occupied the cultivated areas in the later period. Fig. 1 reveals the two primary types of nonagriculturalization: 1) cultivated lands close to the residential areas, primarily converted into residential zones as shown in Fig. 3(e) and (f) and 2) cultivated lands located in remote regions from residential regions, mainly for the construction of agricultural greenhouses as shown in Fig. 3(g) and (h). The dataset for cultivated land change detection presented in this article not only reduces the omission of manual interpretation but also facilitates large-scale detection of nonagriculturalization, thereby providing an available approach to enhance the efficiency of nonagriculturalization detection.

III. Methodology

This section provides an overview of the overall architecture of SNUNet3+. Initially, the proposed network draws inspiration from UNet3+ and modifies it to a Siamese weight-shared encoder–decoder structure. To address the issues of gradient degradation that may occur during the back-propagation in deep CNNs, a residual network is employed as the encoder, which enhances the network's ability to extract features reliably. Subsequently, the feature enhancement modules, namely, the scSE attention mechanism and the deep supervision module for the supervised hidden layer, are introduced stepwise to improve the detection accuracy.

A. Overview

UNet3+ addresses the limitations of plain skip-connection in UNet and dense connection in UNet++ by redesigning these connections as a new skip connection that effectively captures information from input data at full scales. Medical image organ segmentation experiments have confirmed that UNet3+ exhibits lower model complexity and higher segmentation accuracy than UNet and UNet++ [39]. In the proposed method, UNet3+ is transformed into a Siamese encoder structure while preserving the full-scale style skip connection. The nonagriculturalized targets in the images may have diverse sizes and shapes, and their spectral feature can also differ in different scenes. Full-scale skip connections can capture features at different scales, helping the model to better understand the contextual information in the images, thereby improving the accuracy of cultivated land change detection. This configuration connects the encoders and decoders at different stages, as depicted in Fig. 4. The output features from the Siamese encoder are decoded using a unified decoder structure.

To enhance the generation of precise binary change maps, the scSE module is embedded in the decoder subnetwork to make the network focus on more crucial features. Furthermore, the inclusion of the deep supervision module facilitates the network in learning effective features more easily during the training process, thereby enhancing the learning capacity of the network's hidden layers. Finally, the feature map is partitioned into binary change maps using the sigmoid operation.

Let $I_1, I_2 \in \mathbb{R}^{C\times W\times H}$ represent two images of the same region captured at different time points T_1 (before the change) and T_2 (after the change), respectively, $W \times H$ represents the size of the image, and C denotes the number of channels of the image. As shown in Fig. 5, the process of SNUNet3+ can be summarized as follows.

1) First, images T_1 and T_2 are fed into the feature extractor with shared weights. The feature extractor produces multiscale features denoted by feat_1, feat_2, feat_3, feat_4, and feat_5 by concatenating the extracted features at different scales. Various scale encoders E^i_{T1} and E^i_{T2} ($i = 1$–5) concatenate the extracted T_1 and T_2 features to obtain the multi-scale feature set $F = \{\text{feat}_1, \text{feat}_2, \text{feat}_3, \text{feat}_4, \text{feat}_5\}$.
2) Then, the features of different scales in the feature set F are combined with the decoder through full-scale

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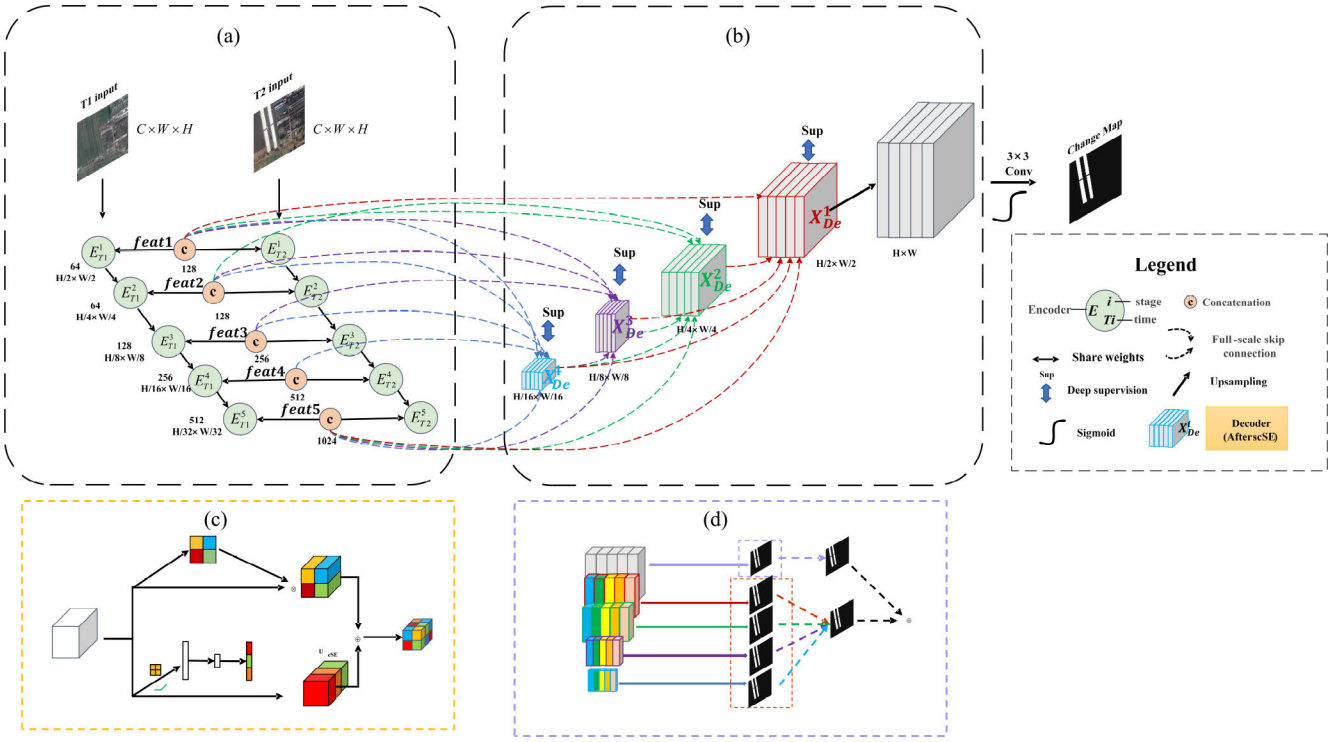


Fig. 4. Overall structural diagram of SNUNet3+. The encoder part of SNUNet3+ is designed as a Siamese structure to extract features from bitemporal images separately. The bitemporal features are concatenated together and then fed into full-scale skip connections. Each decoder is recalibrated by the scSE attention mechanism. The deep supervision module is also embedded into the decoder subnetwork to enable fuller learning of the hidden layer. After the resolution of the feature map is restored to the same as the original image, the prediction map is output by a convolution operation and a Sigmoid operation. (a) Encoder network. (b) Decoder network. (c) scSE attentation mechanism. (d) Deep supervision.

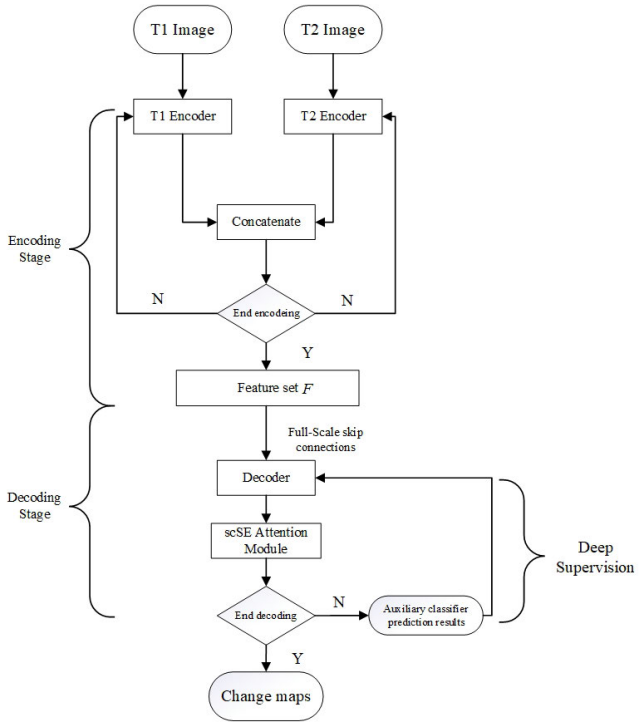


Fig. 5. Flowchart of the change detection in SNUNet3+.

connections. During the decoding stage, each decoder X^i_{De} undergoes calibration through the scSE attention mechanism, enabling it to acquire more discriminative features. A deep supervision module is employed to supervise the intermediate outputs of the decoder and facilitate the learning process and capture information at different levels. The loss function is formulated as follows.

Then, the features of different scales in the feature set F are combined with the decoder through full-scale connections. During the decoding stage, each decoder X^i_{De} undergoes calibration through the scSE attention mechanism, enabling it to acquire more discriminative features. A deep supervision module is employed to supervise the intermediate outputs of the decoder and facilitate the learning process and capture information at different levels. The loss function is formulated using binary cross-entropy (BCE) loss for the final output and focal loss for the intermediate outputs of the decoder, respectively. After computing the losses, they are fused together in a weighted manner to form the overall loss of the network.

3) Finally, after the final upsampling operation, the feature map is restored to the same size as the original image. To obtain the probability map of change detection, the feature map is passed through a convolutional operation followed by a sigmoid activation function.

B. Feature Extractor

As shown in Fig. 4, the encoder is designed as a dual-branch shared-weight convolution with five stages. We attempt to balance the model's parameter count, computational complexity, and the performance of a feature extractor by introducing the ResNet-34 [22] as the backbone of the encoder with

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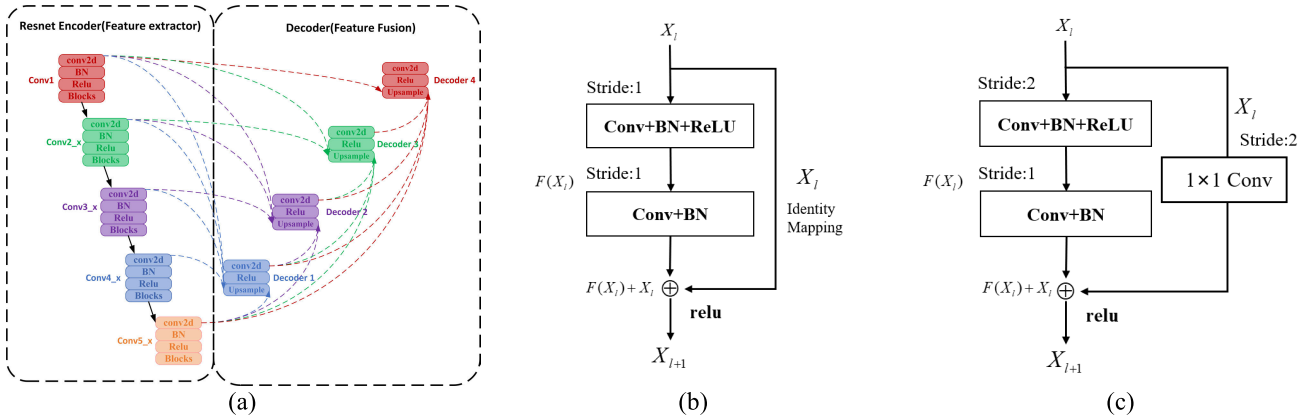


Fig. 6. Structure of basic blocks in ResNet-34. (a) Brief representation of encoder and decoder in the SNUNet3+. (b) General short-circuit connection of basic blocks. (c) 1×1 convolutional short-circuit connection.

the weights pretrained on ImageNet [43]. Fig. 6(a) shows the simplified network framework, omitting the full-scale connection between the encoder–decoder and the decoder. The first layer consists of a 7×7 convolutional layer, which is responsible for extracting low-level texture features. To expand the receptive field, a max pooling layer with a stride of 2 performs downsampling on the original image, reducing its size by half.

The structure of the basic residual block, depicted in Fig. 6, plays a crucial role in the remaining part of the encoder. It consists of four convolutional layers with varying numbers of residual basic blocks. Generally, the basic residual block consists of two weight layers and short-cut connections. The first weight layer combines a 3×3 convolution layer with batch normalization (BN) [44] and the rectified linear unit (ReLU) activation function. The second layer consists of only a convolution layer and BN. The calculation of the residual unit depicted in Fig. 6(c) can be expressed by the following equation:

$$y_l = h(x_l) + F(x_l, W_l) \tag{1}$$

$$x_{l+1} = f(y_l) \tag{2}$$

where x_l and x_{l+1} are the input and output of the residual unit, respectively. $h(x_l) = x_l$ denotes the identity mapping, F denotes the convolution operation of the weight layer, and W_l is the weight of the convolution layer. $f(y_l)$ denotes the ReLU activation function.

In the original ResNet-34, the short-cut connection in the first residual unit of Conv2_x, Conv3_x, and Conv4_x are modified from identity mapping to a 1×1 convolution. This change is made to adjust the number of channels and the size of the feature map

$$y_l = F(x_l, W_l) + F_s(x_l, W_s) \tag{3}$$

where F_s denotes the 1×1 convolution for adjusting the number of channels of shallow layer l and the downsampling operation.

The original ResNet was designed for image classification on the ImageNet dataset. Therefore, the last residual basic block in ResNet is followed by a global average pooling layer and a fully connected layer. As a result, a 1000-D vector (corresponding to the number of categories in ImageNet) is generated. To facilitate upsampling and restore the downsampled features to the original image size, we remove the global average pooling layer and the fully connected layer from ResNet34, creating a variant as a feature extractor in our network. Unlike vgg16 [45], ResNet downsamples the image at each stage of the encoder. As a result, after five rounds of feature extraction, the size of the final feature map is adjusted to $(1/2^5)$ of the original image, i.e., $(H/32) \times (W/32)$. Therefore, after the decoder X_{De}^1 (when the feature map size is $(H/2) \times (W/2)$), an upsampling operation is performed to restore the feature map to the original image size.

Assuming the size of input image pairs is 256×256, the encoder structure, the output size, and the number of residual basic blocks is shown in Table I.

TABLE I
PARAMETERS AND OUTPUT SIZE OF SNUNET3+'S ENCODER

Layer Name	Output Size	33-Layer
Conv1	128×128	7×7, 64, stride2 3×3 max pool, stride2
Conv2_x	64×64	$\begin{bmatrix} 3\times3, 64 \\ 3\times3, 64 \end{bmatrix} \times 3$
Conv3_x	32×32	$\begin{bmatrix} 3\times3, 128 \\ 3\times3, 128 \end{bmatrix} \times 4$
Conv4_x	16×16	$\begin{bmatrix} 3\times3, 256 \\ 3\times3, 256 \end{bmatrix} \times 6$
Conv5_x	8×8	$\begin{bmatrix} 3\times3, 512 \\ 3\times3, 512 \end{bmatrix} \times 3$

C. Spatial and Channel Squeezes and Excitation Module

Fig. 7 illustrates the process of obtaining the third decoder, denoted as X_{De}^3. An example of full-scale skip connection is presented in Fig. 7(a). Let $U \in \mathbb{R}^{C \times W \times H}$ be the feature obtained after full-scale concatenation, where $W \times H$ and C denote the size and the channels number of the feature map, respectively. Before the skip connections, the features from the encoder/decoder are downsampled/upsampled to match the feature resolution size in the decoder. Through the combination of five skip connections, the number of channels

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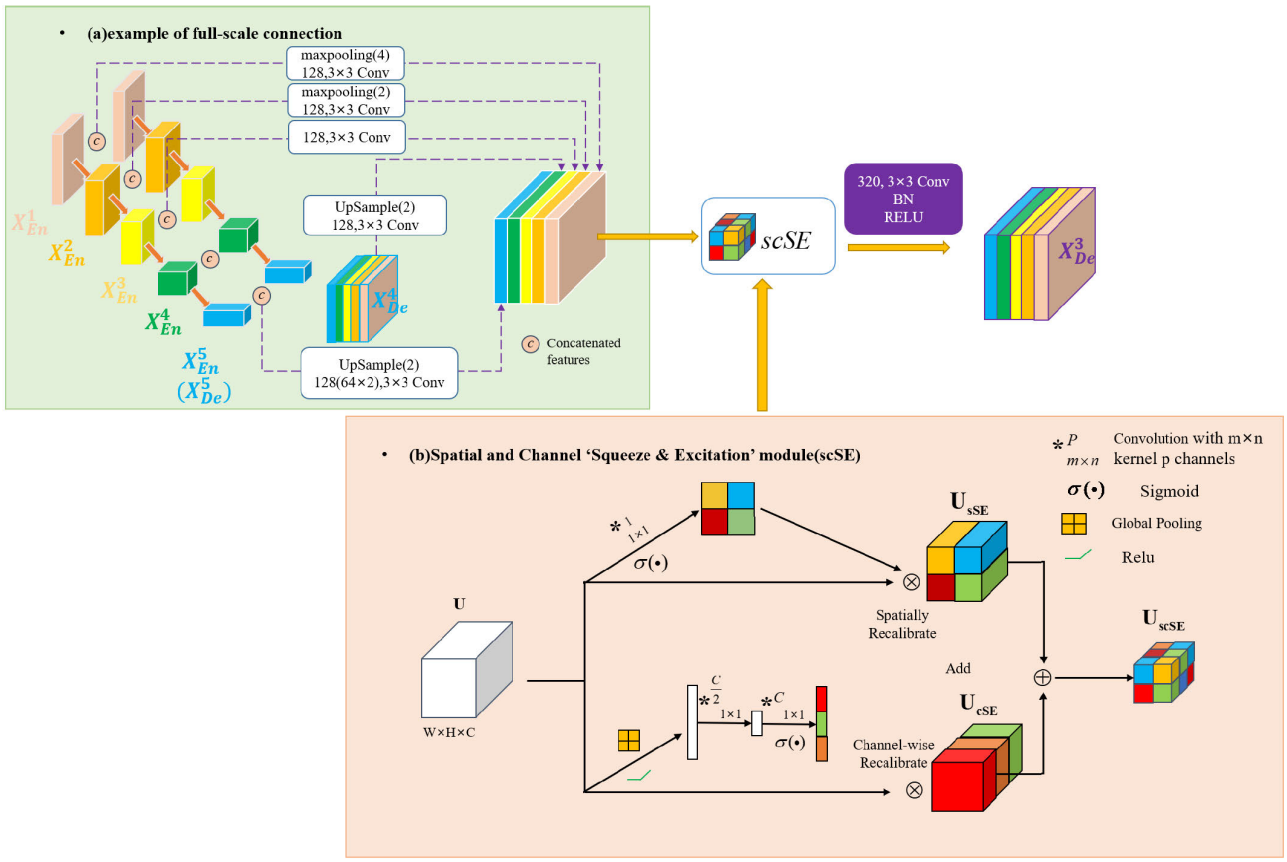


Fig. 7. Description of the construction of the third full-scale decoder X^3_{De}. (a) Example of full-scale connection. (b) Structure of cSE, sSE, and scSE block.

C of the feature, which integrates fine-grained details and coarse-grained semantic information, is calculated by $C = 64 \times 2 \times 5 = 640$. This feature carries a wealth of information, which also contains redundancy. scSE [46] was fused with each decoder to obtain more discriminative features, which abstracts the features into a weight vector in the channel and spatial domains. The features of non-agriculturalized areas may differ at different scales. By introducing the scSE attention mechanism, the model can pay more attention to the target areas based on the features on different scales, thereby capturing more detailed information, which helps to segment nonagriculturalized areas more precisely. Then, scSE is attached to the original spatial and channel features to enhance meaningful features and suppress weaker features. As depicted in Fig. 7(b), scSE consists of two submodules: 1) spatial squeeze and channel excitation block (cSE) and 2) channel squeeze and spatial excitation block (sSE).

*cSE:*The feature map is first passed through the global averaging pooling layer, and the input feature map is considered as a combination of C channels $U = [u_1, u_2, \ldots, u_c], u_i \in \mathbb{R}^{H \times W}$. Equation (4) represents the global averaging pooling of the feature map

$$z = \frac{1}{H \times W} \sum_{i}^{H} \sum_{j}^{W} u_k(i, j). \tag{4}$$

After the pooling operation to generate vector z, the excitation vector $\hat{z}$ in the channel direction is obtained by the following equation:

$$\hat{z} = \sigma(W_1(\delta(W_2 z))) \tag{5}$$

where $W_1 \in \mathbb{R}^{(C/2)\times 1\times 1}$ and $W_2 \in \mathbb{R}^{C\times 1\times 1}$ denote the weights of the two 1×1 convolutions. The ReLU function $\delta(\cdot)$ and the sigmoid function $\sigma(\cdot)$ are used. Finally, the calibrated feature map $\hat{U}_{\text{cSE}}$ is obtained by element-wise multiplication in the channel direction

$$\hat{U}_{\text{cSE}} = U \times \hat{z}. \tag{6}$$

sSE: A $1 \times 1 \times 1$ convolution is applied directly to the feature map U, compressing it along the channel dimension. This operation transforms the shape of U from $[B, C, H, W]$ to $[B, 1, H, W]$, where B represents the batch size, C represents the number of channels, H represents the height, and W represents the width of the feature map. Then, a sigmoid activation function is applied to obtain the spatial attention map. Finally, the spatial attention map is directly applied to the original feature map, completing the spatial information calibration. The calculation method of sSE is represented by the following equation:

$$\hat{U}_{sSE} = \sigma(W_{sq} \times U) \tag{7}$$

where $W_{sq} \in \mathbb{R}^{1\times 1\times 1}$ denotes the weight of the convolution layer and $\sigma(\cdot)$ denotes a sigmoid function to rescale activations to [0, 1].

Through the calculations of cSE and sSE layers, the reshaped features from the channel domain and spatial domain

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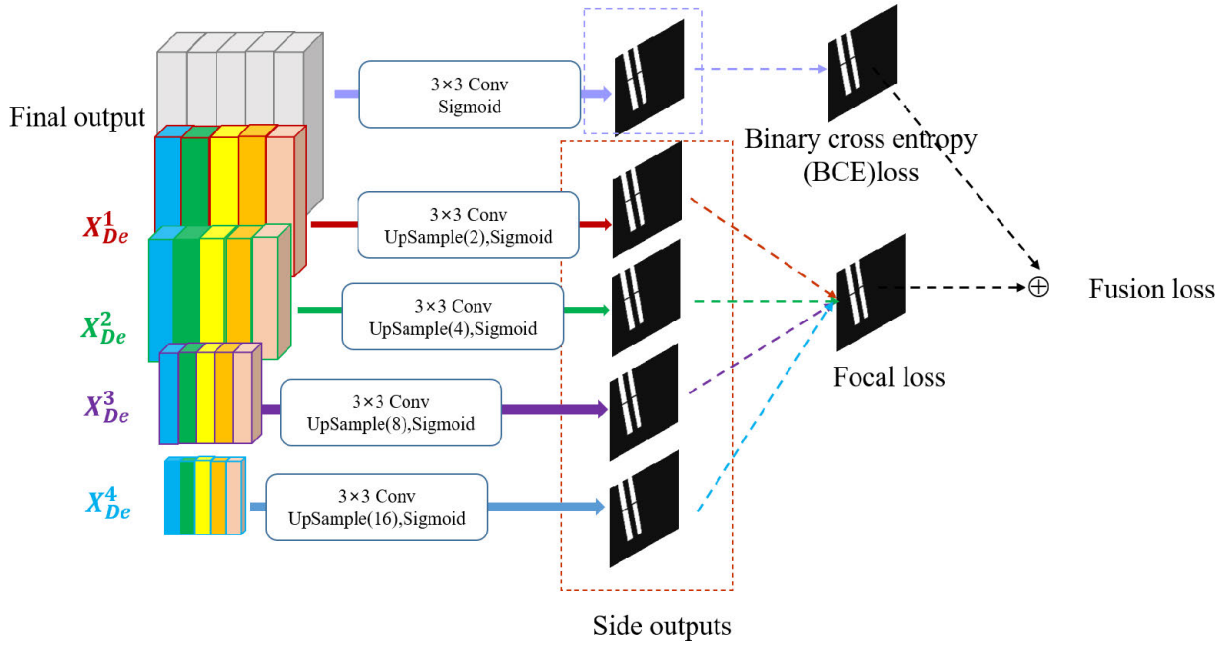


Fig. 8. Flowchart of the deep supervision module. First, calculate the focal losses of the output prediction maps X^1_{De}, X^2_{De}, X^3_{De}, and X^4_{De} and the average of these losses. Subsequently, calculate the BCE loss of the final output of the network. Finally, sum these two losses to determine the total loss for the network.

are summed to obtain the scSE attentional feature $\hat{U}_{\mathrm{scSE}}$, as shown in the following equation:

$$\hat{U}_{\mathrm{scSE}} = U_{\mathrm{cSE}} + U_{\mathrm{sSE}}. \tag{8}$$

D. Deep Supervision Module

To enhance the feature extraction capability of the hidden layers in the network, we introduced a deep supervision module to the network by adding an auxiliary classifier to each decoder of SNUNet3+ as a network branch to supervise the backbone network. Moreover, additional losses are added in the intermediate section of the network.

The cultivated land nonagriculturalization detection task studied in this article typically requires capturing features at different levels, such as texture, shape, and semantics. Through the deep supervision module, the model can be prompted to learn these features at different levels, so that the feature maps can gradually upgrade from low-level features to high-level semantic features and enhance the quality of the detection results. As illustrated in Fig. 8, the decoders $X^1_{\mathrm{De}}, X^2_{\mathrm{De}}, X^3_{\mathrm{De}}, X^4_{\mathrm{De}}$ are first passed through a 3×3 convolutional layer. Then, they undergo upsampling to match the size of the input images. Finally, a sigmoid activation function is applied to the upsampled feature maps, generating a probability map. The side output of each decoder is supervised by assigning a loss function, similar to the final output layer of the network.

We describe the outputs of decoders $X^1_{\mathrm{De}}, X^2_{\mathrm{De}}, X^3_{\mathrm{De}}$, and X^4_{De} as $\mathrm{Out}^n, n = 14$, and the total loss of the decoder's side outputs is calculated by the following equation:

$$L_s = \mathrm{avg}\left(\sum_{n=1}^{4} l(\mathrm{Out}^n)\right) \tag{9}$$

where $l(\cdot)$ is the loss function used by the auxiliary classifier Out^n and avg denotes taking the average.

Assuming that the output of the final layer of the network is denoted by Out_0, the loss of the auxiliary network is obtained and combined with the loss of the final layer $L_0(\mathrm{Out}_0)$ to form the loss function of the whole network

$$\mathrm{Loss} = L_0(\mathrm{Out}_0) + \alpha_t L_s \tag{10}$$

where α_t is the weight factor controlling the importance of the loss of the auxiliary classifier.

E. Loss Function

Illegal cultivated land occupations are gradually decreasing given the implementation of strict regulations by government agencies. Consequently, the PX-CLCD dataset used in this article contains relatively small instances, where cultivated land has changed compared with the overall cultivated land area. Specifically, the number of unchanged samples significantly outweighs the number of changed samples in the PX-CLCD dataset. Focal loss [47] is introduced in the deep supervision module to help the network pay more attention to the minority class samples; in this case, the "changed" samples during the training process and to alleviate the imbalance between positive and negative samples. The focal loss is computed for each output layer in the deep supervision module according to the following equation:

$$L_{\mathrm{Fl}} = -\alpha(1 - p_t)^{\gamma} \log(p_t) \tag{11}$$

where p_t represents the predicted probability of a certain class by the model, α balances the importance of positive/negative example, and γ is a tunable focusing parameter.

BCE loss is usually applied in binary classification problems. For batch data (x_n, y_n) containing N samples, x_n denotes the prediction map, and y_n denotes the ground truth. The BCE

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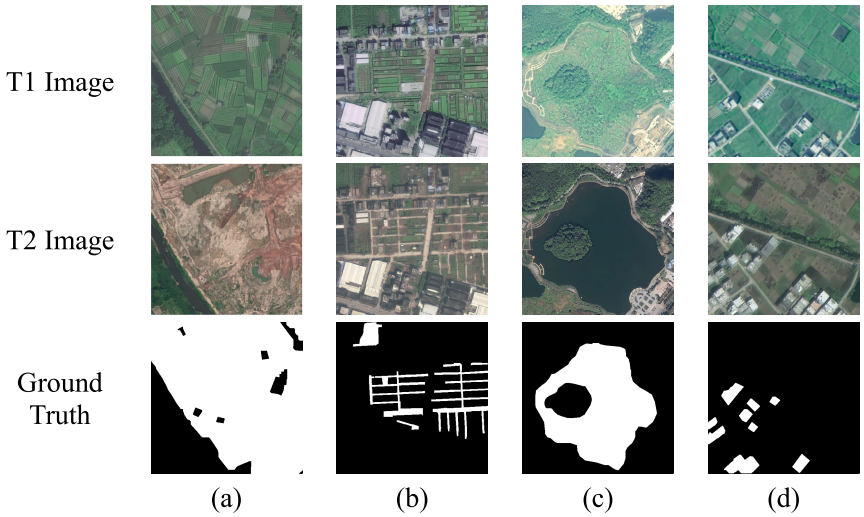


Fig. 9. Example images of CLCD dataset (512 × 512). Each column represents a different change category. (a) Bare land. (b) Roads. (c) Lakes. (d) Buildings.

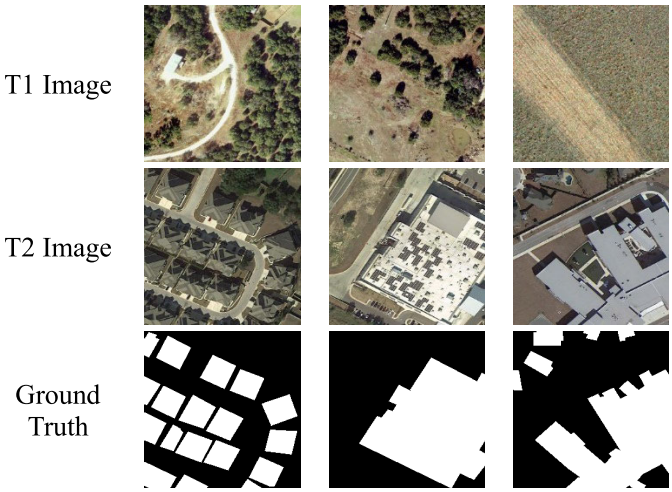


Fig. 10. Example images of LEVIR-CD dataset (256 × 256).

loss is applied to the final output of SNUNet3+, as shown in the following equations:

$$L_{\mathrm{BCE}} = \frac{1}{N} = \sum_{n=1}^{N} l_n \tag{12}$$

$$l_n = -[y_n \log x_n + (1 - y_n) \log(1 - x_n)] \tag{13}$$

where l_n is the loss corresponding to the nth sample.

Finally, the total loss of the overall network is derived from the combination of the final layer loss L_{BCE} and the auxiliary classifier loss L_{Fl}

$$\mathrm{Loss} = \mathrm{avg}\left(\sum_{i=1}^{4} L_{\mathrm{Fl}}(\mathrm{Out}^n)\right) + \alpha_t L_{\mathrm{BCE}}(\mathrm{Out}_0) \tag{14}$$

where α_t is the weight factor that controls the importance of the loss of the auxiliary classifier. αt is calculated as follows:

$$\alpha_t \leftarrow \alpha_t \times (1 - t/N) \tag{15}$$

where t denotes the current epoch and N denotes the total number of epochs.

IV. Results and Analysis

A. Experimental Datasets

To verify the performance of the proposed method in this research, three experimental datasets were selected: CLCD, LEVIR-CD, and PX-CLCD datasets.

1) The CLCD dataset [38] consists of 600 pairs of cultivated land change sample images, of which 320 pairs are used for training, 120 for validation, and 120 for testing. The bitemporal images in the CLCD were acquired by the Gaofen-2 satellite in Guangdong Province, China, in 2017 and 2019, respectively, with spatial resolutions ranging from 0.5 to 2 m. Each set of samples consists of two 512 × 512 images and a corresponding binary label of cropland change. In the CLCD dataset, the primary types of changes labeled include buildings, roads, lakes, and bare land. It is a challenging dataset for cultivated land change detection and can effectively evaluate the performance of SNUNet3+ for cultivated land change detection. The main types of changes in CLCD are summarized and shown in Fig. 9.

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2) The LEVIR-CD [32] dataset contains 637 pairs of Google Earth images with an ultrahigh resolution of 0.5 m. Each pair of images in the dataset is 1024 × 1024 pixels. This dataset covers several cities in TX, USA, and includes image pairs with a period ranging from 5 to 14 years. These images capture significant land use changes in the study area, particularly the increase in buildings. The LEVIR-CD dataset encompasses various types of buildings, such as villas, high-rise apartments, small garages, and large warehouses. The fully labeled LEVIR-CD dataset contains a total of 31 333 instances of building change. The dataset is divided into a training set, a validation set, and a test set in the ratio of 7:2:1. It serves as a benchmark dataset for evaluating the change detection algorithm and objectively assessing its performance improvement. The main types of changes in CLCD are summarized and shown in Fig. 10.
3) PX-CLCD dataset focuses specifically on the changes occurring in cultivated land with complex change scenarios and moderate data volume, which is expected to achieve better detection performance.

B. Comparison With State-of-the-Art Methods

To demonstrate the performance improvement of the proposed method, six classic baseline methods for change detection are selected for comparative experiments. The selected baseline methods are as follows.

1) *FC-EF [28]:* This method is based on the UNet structure. It concatenates two images in the channel direction, treating them as multispectral images with different channels. The encoder incorporates feature fusion with the corresponding scale decoder through skip connections.
2) *FC-Siam-conc [28]:* A dual-branch encoder network with shared weights is constructed in this method. Each branch performs independent feature extraction on the diachronic input image. FC-Siam-conc concatenates the features from the dual-branch encoder and establishes skip connections with the corresponding decoder features.
3) *FC-Siam-diff [28]:* Similar to FC-Siam-conc, this method also constructs a Siamese encoder network with shared weights. However, FC-Siam-diff first calculates the absolute difference between the feature maps of the two encoder branches and then establishes skip connections with the corresponding decoder layers.
4) *SNUNet [34]:* This method designs a densely connected Siamese network inspired by the densely connected UNet (UNet++). In addition, it proposes an integrated channel attention module (ECAM) to supervise the outputs of different semantic levels within the middle section of SNUNet, which contributes to the final classification.
5) *STANet [32]:* This method introduces ResNet-18 as the feature extraction network backbone. STANet integrates a spatial–temporal self-attention mechanism that captures spatial–temporal dependencies at various scales.
6) *MSCANet [38]:* This method is a CNN-transformer network with multiscale context aggregation. It employs CNN for original image feature extraction, a multiscale context-integrated transformer for encoding and decoding the features, and a multibranch CNN classifier to supervise the output from different network branches and obtain change maps.
7) *DMINet [48]:* This method is a dual-branch multilevel intertemporal network. unifying self-attention and cross-attention in a single module, we present an intertemporal joint-attention block to steer the global feature distribution of each input, motivating information coupling between intralevel representations and meanwhile suppressing the task-irrelevant interferences.
8) *A2Net [49]:* This method is a novel lightweight network, which identifies changes based on the features extracted by mobile networks via progressive feature aggregation and supervised attention.

C. Parameters and Evaluation Metrics

In this article, all experiments are implemented using PyTorch [50]. The model training is accelerated using NVIDIA RTX3090 with 24G video memory for 100 epochs. Due to the GPU memory limitation, we crop the original size of 1024 × 1024 LEVIR-CD and the original size of 512 × 512 image of CLCD to 256 × 256 to obtain better training metrics. The batch sizes of CLCD, LEVIR-CD, and PX-CLCD are all set to 8. The initial learning rate of the Adam optimizer is set to 1e−4 with 1% decay every 20 epochs. In Focal Loss, the hyperparameters α and γ are set to 0.5 and 2, respectively. The KaiMing initialization algorithm is used to initialize the weights of the convolutional layers in the model.

To evaluate the performance of the models in the experiment, we employed five classical evaluation metrics derived from the confusion matrix: precision, recall, $F1$-score, accuracy, and intersection over union (IoU). Precision measures the model's accuracy in predicting positive samples, recall measures the model's ability to identify positive samples correctly, and the $F1$-score is the harmonic mean of the precision and recall, providing a balanced measure of the model's performance. Accuracy measures the overall correctness of the model's predictions by calculating the ratio of correctly classified instances to the total number of instances. IoU quantifies the similarity between predicted and true values. These metrics can be computed as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{16}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{17}$$

$$F_1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{18}$$

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}} \tag{19}$$

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \tag{20}$$

where TP, FP, FN, and TN are the numbers of true positives, false positives, false negatives, and true negatives, respectively.

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TABLE II
EXPERIMENTAL RESULTS ON CLCD DATASET

Method	Pr(%)	Re(%)	F1(%)	Iou(%)	OA(%)
FC-EF	60.7	54.8	56.58	40.48	93.99
FC-Siam-conc	54.53	61.56	57.48	41.48	93.69
FC-Siam-diff	66.79	51.71	57.05	40.78	94.72
SNUNet	62.82	56.48	58.96	42.55	94.65
STANet	66.64	69.18	66.70	50.77	95.29
MSCANet	73.75	68.45	71.74	55.04	95. 84
DMINet	58.08	52.21	53.26	37.78	94.63
A2Net	70.26	58.32	62.65	47.12	96.06
SNUNet3+	**78.50**	**68.95**	**72.90**	**58.11**	**96.37**

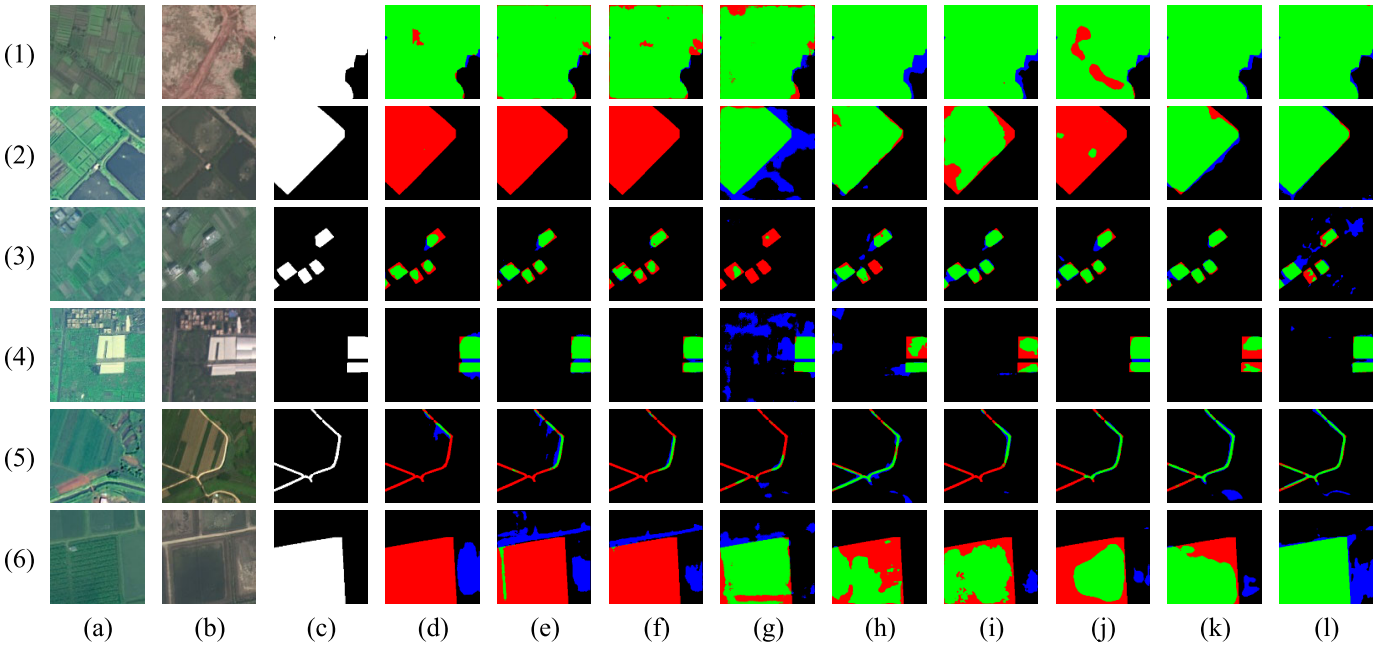


Fig. 11. Visualization comparison of different methods on CLCD. (a) T1 image. (b) T2 image. (c) Ground truth. (d) FC-EF. (e) FC-Siam-conc. (f) FC-Siam-diff. (g) SNUNet. (h) STANet. (i) MSCANet. (j) DMINet. (k) A2Net. (l) SNUNet3+. Green indicates the true positive, blue indicates the false positive, red indicates the false negative, and black the indicates true negative.

D. Experimental Results on the CLCD dataset

The evaluation results of the proposed method and other baseline methods on the CLCD dataset are presented in Table II, which shows that SNUNet3+ outperforms the other baseline methods in terms of all evaluation metrics. Specifically, SNUet3+ achieves a higher $F1$-score compared with FC-EF, FC-Siam-conc, FC-Siam-diff, SNUNet, STANet, MSCANet, DMINet, and A2Net with improvements of 16.32%, 15.42%, 15.85%, 13.94%, 6.2%, 1.16%, 19.64%, and 10.25%, respectively. The superior performance of SNUNet3+ can be attributed to utilizing full-scale skip connections, the scSE attention mechanism, and the deep supervision module. Despite MSCANet incorporating a transformer as a complement to CNN to address the limitations of the local receptive field, its performance in CLCD is still slightly inferior to that of SNUNet3+. The performance of FC-EF is the lowest among all methods, indicating that the early fusion methods suffer from inadequate mining of deep feature information in CLCD.

Fig. 11 shows examples of the results obtained from SNUNet3+ and the baseline methods after change detection using CLCD. As shown in row (1) of Fig. 11, all methods successfully identify regions with significant and substantial changes. However, for regions with semantic changes, such as the conversion of cultivated land to other land types in rows (2) and (5), FC-EF, FC-Siam-conc, and FC-Siam-diff fail to detect these changes effectively, resulting in a high false negative rate. In contrast, our method excels at detecting such changes while preserving the integrity of change boundaries. Furthermore, as shown in rows (4) and (5), our method achieves the most accurate extraction results for subtle change targets, such as small buildings and roads.

E. Experimental Results on the LEVIR-CD dataset

The various evaluation metrics obtained after change detection on CLCD by the proposed method and other baseline methods are shown in Table III. FC-EF exhibited the lowest $F1$ score of 75.33% among the baseline methods. Except for FC-EF, all other methods performed better on LEVIR-CD. SNUNet3+ achieved the highest $F1$ score of 90.36% and IoU of 82.49% on LEVIR-CD, indicating its superior performance in these important metrics.

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TABLE III
EXPERIMENTAL RESULTS ON LEVIR-CD DATASET

Method	Pre(%)	Rec(%)	F1(%)	Iou(%)	OA(%)
FC-EF	76.73	74.89	75.33	61.91	97.90
FC-Siam-conc	87.68	84.21	85.69	76.15	98.84
FC-Siam-diff	87.84	79.32	83.00	72.97	98.7
SNUNet	89.24	84.50	86.34	77.59	98.97
STANet	91.27	84.64	87.38	78.30	99.09
MSCANet	90.87	87.79	87.94	80.68	98.93
DMINet	91.27	82.25	86.07	77.11	98.95
A2Net	90.38	85.41	87.36	79.21	**99.09**
SNUNet3+	**92.38**	**88.50**	**90.36**	**82.49**	99.07

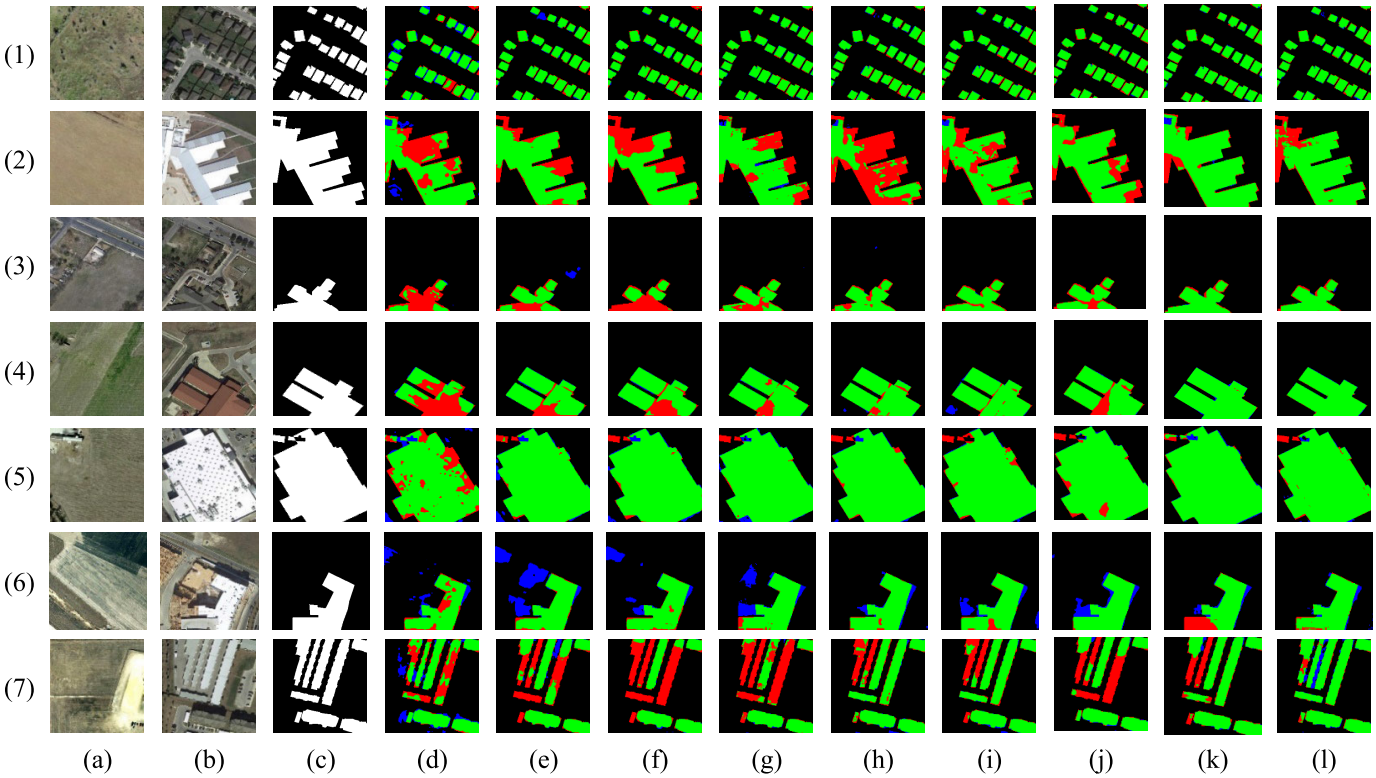


Fig. 12. Visualization comparison of different methods on the LEVIR-CD dataset. (a) T1 image. (b) T2 image. (c) Ground truth. (d) FC-EF. (e) FC-Siam-conc. (f) FC-Siam-diff. (g) SNUNet. (h) STANet. (i) MSCANet. (j) DMINet. (k) A2Net. (l) SNUNet3+. Green indicates the true positive, blue indicates the false positive, red indicates the false negative, and black indicates the true negative.

Fig. 12 presents the example results obtained from change detection on the LEVIR-CD dataset by SNUNet3+ and various baseline methods. Regarding small buildings, which have well-defined boundaries and are prevalent in the LEVIR-CD, all methods can accurately identify these small-scale buildings that have changed. However, for large-scale buildings with irregular shapes, it was challenging for all benchmark methods to accurately depict completely changing objects because of the limitation in sample quantity and the feature extraction capabilities of the models, as depicted in row (2) of Fig. 12. Benefiting from UNet3+'s ability to capture spatial boundary information from the original image and the integration of the scSE attention mechanism in the decoder, the proposed method generated relatively complete boundaries in the binary change map compared with all baseline methods, as shown in line (2) of Fig. 12. Furthermore, as evidenced by rows (3), (4), and (6) in Fig. 12, SNUNet3+ produced the most comprehensive binary change maps, accurately depicting the boundaries and preserving detailed internal information of the change objects.

F. Experimental Results on the PX-CLCD Dataset

The evaluation metrics obtained by the proposed method and other baseline methods on PX-CLCD are shown in Table IV. Unlike the results on CLCD and LEVIR-CD, FC-EF outperforms FC-Siam-conc, and FC-Siam-diff on PX-CLCD, with a higher $F1$ score of approximately 10% compared with the other two methods. This observation highlights the importance of a powerful feature extraction capability in the encoder for Siamese change detection models. The attention mechanism module embedded in SNUNet, STANet, MSCANet, and SNUNet3+ contributes to their strong performance on PX-CLCD. In terms of $F1$-Score, SNUNet3+ demonstrates improvements of 9.18%, 22.69%, 22.37%, 1.4%,

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TABLE IV
EXPERIMENTAL RESULTS ON PX-CLCD DATASET

Method	Pre(%)	Rec(%)	F1(%)	Iou(%)	OA(%)
FC-EF	89.68	85.56	87.46	78.07	97.01
FC-Siam-conc	74.98	73.69	73.95	59.24	93.57
FC-Siam-diff	76.71	72.72	74.27	59.64	93.80
SNUNet	95.28	95.33	95.24	91.03	98.85
STANet	93.26	94.49	93.82	88.46	98.46
MSCANet	94.61	92.93	93.68	88.26	98.40
DMINet	94.49	92.26	93.26	87.48	98.33
A2Net	96.53	95.99	96.22	92.78	99.06
SNUNet3+	**96.60**	**96.79**	**96.64**	**93.61**	**99. 19**

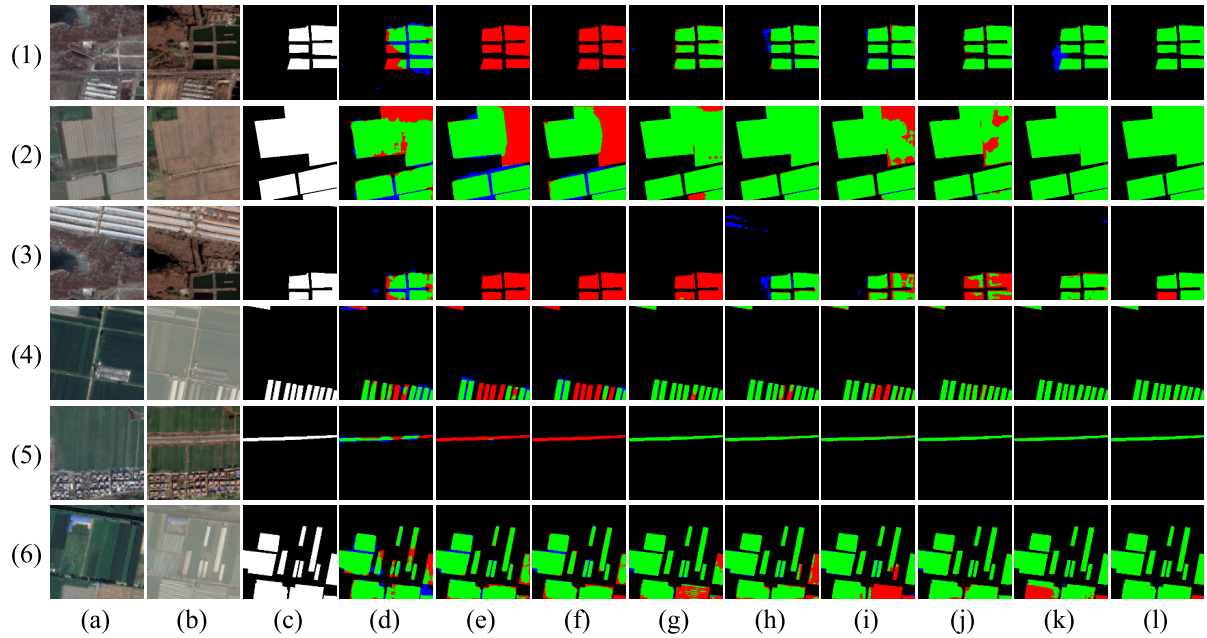


Fig. 13. Visualization comparison of different methods on the LEVIR-CD dataset. (a) T1 image. (b) T2 image. (c) Ground truth. (d) FC-EF. (e) FC-Siam-conc. (f) FC-Siam-diff. (g) SNUNet. (h) STANet. (i) MSCAnet. (j) DMINet. (k) A2Net. (l) SNUNet3+. Green indicates the true positive, blue indicates the false positive, red indicates the false negative, and black indicates the true negative.

2.82%, 2.96%, 3.38%, and 0.42% over FC-EF, FC-Siam-conc, FC-Siam-diff, SNUNet, STANet, MSCANet, DMINet, and A2Net, respectively, emerging as the top-performing method.

Fig. 13 presents the example images depicting the change detection results on PX-CLCD using SNUNet3+ and each baseline method. FC-EF achieves higher accuracy and recalls than FC-Siam-conc and FC-Siam-diff, successfully detecting more change features. Regarding complex scenes in the PX-CLCD dataset, SNUNet, STANet, and MSCANet can identify various changing objects with different scales. These methods have made improvements in their encoders and feature enhancement strategies. Overall, SNUNet3+ outperforms other baseline methods, as evident from its visualization results in Fig. 13, displaying more accurate and complete change detection, which aligns with its highest $F1$ score obtained in the evaluation.

V. DISCUSSION

A. Ablation Study

SNUNet3+ is an extension of UNet3+ that incorporates the scSE attention mechanism and the deep supervision module within the decoder subnetwork. To evaluate the effectiveness of scSE and the deep supervision module, four ablation experiments were performed on the CLCD and LEVIR-CD datasets: 1) "Base" represents the siamese UNet3+ baseline without the scSE attention mechanism and the deep supervision module; 2) "Base + scSE" represents the adoption of the scSE module to the baseline model; 3) "Base + DS" represents the adoption of the deep supervision module to the baseline model; and 4) "SNUNet3+" represents the incorporation of the scSE module and deep supervision module into the baseline method. The hyperparameters were set identically for these four experiments. Fig. 14 displays the plots of the four networks on the CLCD, LEVIR-CD, and PX-CLCD datasets obtained from the validation set during model training. As shown in Fig. 14(a), the $F1$ score curve of SNUNet3+ exhibits the highest value among the four methods during the final stage of training, whereas Fig. 14(b) demonstrates the fastest convergence rate, indicating its superior generalization ability.

Moreover, "Base + scSE" in Fig. 14(b) and "Base + DS" in Fig. 14(a) and (b) converge slightly faster compared with "Base."

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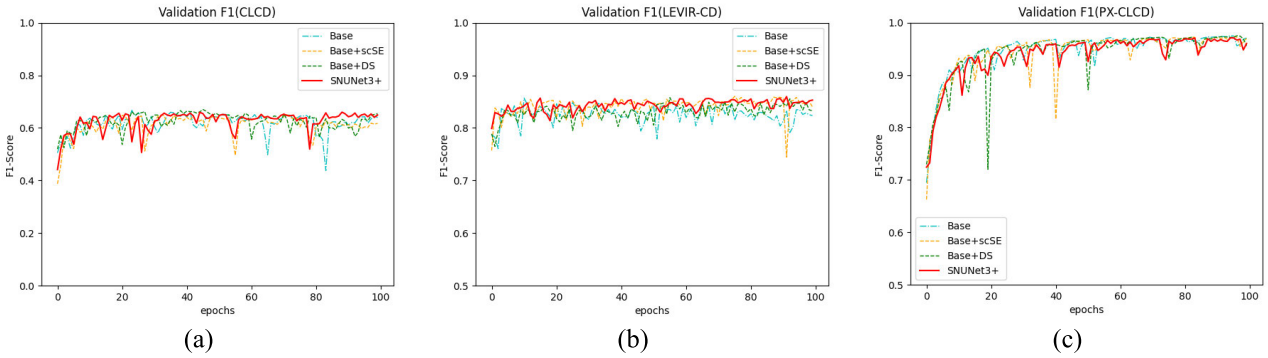


Fig. 14. Comparison of validation curves on CLCD and LEVIR datasets. (a) CLCD. (b) LEVIR-CD. (c) PX-CLCD.

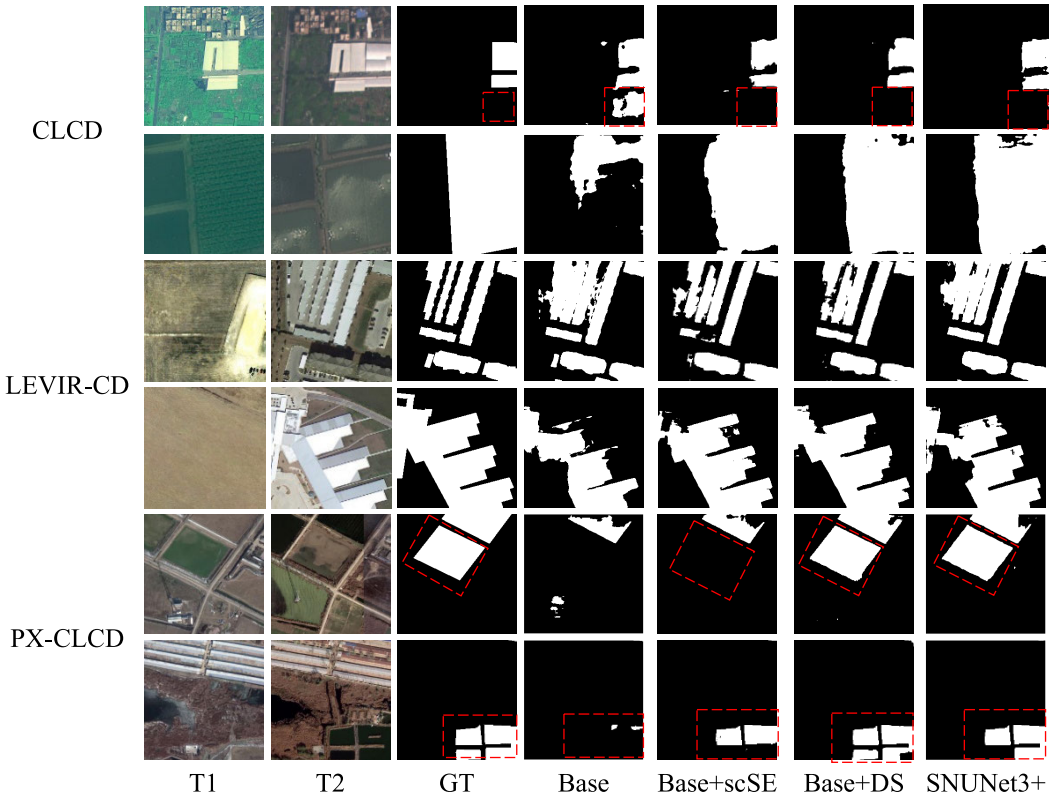


Fig. 15. Visualization of ablation experimental results.

Fig. 15 displays the change maps of the four ablation experiments, which shows that the change maps generated by "Base" exhibit inaccurate boundaries and incomplete internal details, even fail to detect some of the changed areas on the PX-CLCD dataset, indicating the worst quality among the four methods. On the contrary, SNUNet3+ produces change maps with clearer object boundaries and more complete internal details, attributed to including the scSE and deep supervision modules within the network.

Regarding quantitative evaluation metrics, the performance of the baseline model is improved by the addition of scSE and DS, as shown in Table V. The utilization of scSE to generate more discriminative feature maps improved the $F1$ score and IoU by 1.82% and 1.82% on the LEVIR-CD dataset, by 1.6% and 1.67% on the CLCD dataset and by 0.16% and 0.31% on the PX-CLCD dataset, respectively, indicating the beneficial impact of the attention mechanism on change detection. Furthermore, after adding the deep supervision, $F1$ and IoU improved by 3.66% and 3.82% on the CLCD dataset, by 0.54% and 0.2% on the LEVIR-CD dataset and 0.29% and 0.55% on the PX-CLCD dataset, respectively. Combined with the curves of "base" and "Base + ds" in Fig. 14, the deep supervision module has the effect of assisting the learning ability of the hidden layer. Finally, when utilizing scSE and the deep supervision module in SNUNet3+, the $F1$ score and IoU achieve the highest improvements compared with the original model by 4.99% and 3.82% in CLCD, by 3.6% and 3.56% in LEVIR-CD, and by 0.7% and 1.35% on the PX-CLCD dataset, respectively.

B. Model Efficiency

As for the comparison of model efficiency, Table VI exhibits the number of model parameters (Params.), the floating-point

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TABLE V
ABLATION STUDY ON SNUNET3+

Method	CLCD		LEVIR-CD		PX-CLCD	
	F1	Iou	F1	Iou	F1	Iou
Base	67.91	52.49	86.76	78.93	95.94	92.26
Base+scSE	69.51	54.16	88.58	80.75	96.10	92.57
Base+DS	71.57	56.31	87.30	79.13	96.23	92.81
SNUNet3+	72.90	58.11	90.36	82.49	96.64	93.61

TABLE VI
MODEL EFFICIENCY OF DIFFERENT METHODS

Models	Params(M)	FLOPs(G)	Backbones
FC-EF	1.239	3.213	-
FC-Conc	1.546	5.331	-
FC-Diff	1.350	4.727	-
STANet	12.211	12.710	ResNet18
SNUNet	12.035	54.833	-
MSCANet	16.420	14.771	ResNet18
DMINet	6.242	14.551	ResNet18
A2Net	3.780	3.048	MobileNetV2[51]
SNUNet3+	47.452	164.234	ResNet34

operations per second (FLOPs), and the backbones of different models.

As shown in Table VI, the number of parameters and FLOPs of our model are the largest among all the baseline methods. A large number of parameters are generated due to the fact that we introduced a pretrained ResNet34 as the backbone and designed the fully connected encoder that receives the bitemporal images. Although the proposed method outperforms the SOTA methods in terms of evaluation metrics on change detection experiments, the number of parameters and the required training time are not dominant. Therefore, there is still a need to investigate some lightweight architecture to exploit the potential for acceleration.

VI. CONCLUSION

With the progress in artificial intelligence and RS image change detection, the focus of research has shifted toward applying DL techniques for change detection, particularly in land use/cover change. To address the limitations of previous approaches, a novel method called SNUNet3+ is proposed in this research. SNUNet3+ combines the full-scale connected UNet with a Siamese network and incorporates the scSE and deep supervision modules to detect changes in cultivated land using high-resolution images. The Resnet-34 is employed as the network backbone to ensure the stability of the network and mitigate issues like network degradation and gradient disappearance. In addition, the scSE attention mechanism is embedded in the decoder subnetwork to enhance the feature map and reduce the influence of irrelevant information. Moreover, a deep supervision module is introduced to supervise the side outputs of the decoder, facilitating the hidden layers to learn more informative features. SNUNet3+ improved by 4.99%, 3.6%, and 0.7% compared with UNet3+ without scSE and DS in terms of F1 on the CLCD, LEVIR-CD, and PX-CLCD datasets, respectively.

DL models rely on large data to drive their tasks. To achieve the task of cultivated land change detection, a high-resolution image cultivated land change detection dataset PX-CLCD was produced using Pei County, Jiangsu Province as the study area in this research, which achieves the metrics of existing change detection datasets in terms of spatial resolution and data volume.

The method proposed in this article was also applied to the CLCD dataset, LEVIR-CD, and PX-CLCD datasets with other baseline methods, and the experimental results indicated that SNUNet3+ outperforms the other methods on all three datasets. From the change detection results, the change objects extracted by SNUNet3+ can maintain relatively good boundary accuracy and internal information integrity. The combined quantitative and visualization results can show that the combination of full-scale skip connections, attention mechanism, and deep supervision module used in this article has a certain enhancement effect on the change detection task based on DL.

REFERENCES

[1] Z. Huang, X. Du, and C. S. Z. Castillo, "How does urbanization affect farmland protection? Evidence from China," *Resour., Conservation Recycling*, vol. 145, pp. 139–147, Jun. 2019.
[2] Y. Zhou, X. Li, and Y. Liu, "Cultivated land protection and rational use in China," *Land Use Policy*, vol. 106, Jul. 2021, Art. no. 105454.
[3] Y. Chen, S. Wang, and Y. Wang, "Spatiotemporal evolution of cultivated land non-agriculturalization and its drivers in typical areas of Southwest China from 2000 to 2020," *Remote Sens.*, vol. 14, no. 13, p. 3211, Jul. 2022.
[4] K. D. Mello et al., "Multiscale land use impacts on water quality: Assessment, planning, and future perspectives in Brazil," *J. Environ. Manage.*, vol. 270, Sep. 2020, Art. no. 110879.
[5] P. Ghamisi et al., "Multisource and multitemporal data fusion in remote sensing: A comprehensive review of the state of the art," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 1, pp. 6–39, Mar. 2019.
[6] Z. Lv, T. Liu, J. A. Benediktsson, and N. Falco, "Land cover change detection techniques: Very-high-resolution optical images: A review," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 1, pp. 44–63, Mar. 2022.
[7] Q. Zhu et al., "Land-use/land-cover change detection based on a Siamese global learning framework for high spatial resolution remote sensing imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 184, pp. 63–78, Feb. 2022.
[8] S. W. Wang, B. M. Gebru, M. Lamchin, R. B. Kayastha, and W.-K. Lee, "Land use and land cover change detection and prediction in the Kathmandu district of Nepal using remote sensing and GIS," *Sustainability*, vol. 12, no. 9, p. 3925, May 2020.
[9] P. L. Rosin and T. J. Ellis, "Image difference threshold strategies and shadow detection," in *Proc. BMVC*, vol. 95, 1995, pp. 347–356.
[10] L. T. Luppino, F. M. Bianchi, G. Moser, and S. N. Anfinsen, "Unsupervised image regression for heterogeneous change detection," 2019, *arXiv:1909.05948*.
[11] W. J. Todd, "Urban and regional land use change detected by using Landsat data," *J. Res. U.S. Geol. Surv.*, vol. 5, no. 5, pp. 529–534, 1977.
[12] W. A. Malila, "Change vector analysis: An approach for detecting forest changes with Landsat," in *Proc. LARS Symposia*, 1980, p. 385.
[13] H. Abdi and L. J. Williams, "Principal component analysis," *Wiley Interdiscipl. Rev., Comput. Statist.*, vol. 2, no. 4, pp. 433–459, Jul. 2010.
[14] A. A. Nielsen, "The regularized iteratively reweighted MAD method for change detection in multi- and hyperspectral data," *IEEE Trans. Image Process.*, vol. 16, no. 2, pp. 463–478, Feb. 2007.
[15] R. Peiman, "Pre-classification and post-classification change-detection techniques to monitor land-cover and land-use change using multi-temporal Landsat imagery: A case study on Pisa Province in Italy," *Int. J. Remote Sens.*, vol. 32, no. 15, pp. 4365–4381, Aug. 2011.
[16] C. Huang et al., "Use of a dark object concept and support vector machines to automate forest cover change analysis," *Remote Sens. Environ.*, vol. 112, no. 3, pp. 970–985, Mar. 2008.

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[17] J. Im and J. Jensen, "A change detection model based on neighborhood correlation image analysis and decision tree classification," *Remote Sens. Environ.*, vol. 99, no. 3, pp. 326–340, Nov. 2005.

[18] W. Feng, H. Sui, J. Tu, W. Huang, and K. Sun, "A novel change detection approach based on visual saliency and random forest from multi-temporal high-resolution remote-sensing images," *Int. J. Remote Sens.*, vol. 39, no. 22, pp. 7998–8021, Nov. 2018.

[19] H. Jiang et al., "A survey on deep learning-based change detection from high-resolution remote sensing images," *Remote Sens.*, vol. 14, no. 7, p. 1552, Mar. 2022.

[20] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3431–3440.

[21] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. 18th Int. Conf. Med. Image Comput. Comput.-Assist. Intervent. (MICCAI)*, Munich, Germany. Cham, Switzerland: Springer, Oct. 2015, pp. 234–241.

[22] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[23] M. Zhang and W. Shi, "A feature difference convolutional neural network-based change detection method," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 10, pp. 7232–7246, Oct. 2020.

[24] R. Liu, M. Kuffer, and C. Persello, "The temporal dynamics of slums employing a CNN-based change detection approach," *Remote Sens.*, vol. 11, no. 23, p. 2844, Nov. 2019.

[25] E. B. Alexakis and C. Armenakis, "Evaluation of UNet and UNet++ architectures in high resolution image change detection applications," *Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 43, pp. 1507–1514. Aug. 2020.

[26] J. Bromley, I. Guyon, Y. LeCun, E. Säckinger, and R. Shah, "Signature verification using a 'Siamese' time delay neural network," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 6, 1993, pp. 669–688.

[27] S. Zagoruyko and N. Komodakis, "Learning to compare image patches via convolutional neural networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 4353–4361.

[28] R. C. Daudt, B. Le Saux, and A. Boulch, "Fully convolutional Siamese networks for change detection," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2018, pp. 4063–4067.

[29] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7132–7141.

[30] C.-Y. Lee, S. Xie, P. Gallagher, Z. Zhang, and Z. Tu, "Deeply-supervised nets," in *Proc. Artif. Intell. Statist.*, 2015, pp. 562–570.

[31] C. Zhang et al., "A deeply supervised image fusion network for change detection in high resolution bi-temporal remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 166, pp. 183–200, Aug. 2020.

[32] H. Chen and Z. Shi, "A spatial–temporal attention-based method and a new dataset for remote sensing image change detection," *Remote Sens.*, vol. 12, no. 10, p. 1662, 2020.

[33] J. Chen et al., "DASNet: Dual attentive fully convolutional Siamese networks for change detection in high-resolution satellite images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 1194–1206, 2021.

[34] S. Fang, K. Li, J. Shao, and Z. Li, "SNUNet-CD: A densely connected Siamese network for change detection of VHR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[35] H. Chen, Z. Qi, and Z. Shi, "Remote sensing image change detection with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2021, Art. no. 5607514.

[36] W. G. C. Bandara and V. M. Patel, "A transformer-based Siamese network for change detection," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2022, pp. 207–210.

[37] W. Li, L. Xue, X. Wang, and G. Li, "ConvTransNet: A CNN-transformer network for change detection with multi-scale global-local representations," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5610315.

[38] M. Liu, Z. Chai, H. Deng, and R. Liu, "A CNN-transformer network with multiscale context aggregation for fine-grained cropland change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 4297–4306, 2022.

[39] H. Huang et al., "UNet 3+: A full-scale connected UNet for medical image segmentation," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, May 2020, pp. 1055–1059.

[40] S. Tian, A. Ma, Z. Zheng, and Y. Zhong, "Hi-UCD: A large-scale dataset for urban semantic change detection in remote sensing imagery," 2020, *arXiv:2011.03247*.

[41] S. Ji, S. Wei, and M. Lu, "Fully convolutional networks for multisource building extraction from an open aerial and satellite imagery data set," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 574–586, Jan. 2019.

[42] L. Shen et al., "S2Looking: A satellite side-looking dataset for building change detection," *Remote Sens.*, vol. 13, no. 24, p. 5094, Dec. 2021.

[43] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," *Commun. ACM*, vol. 60, no. 6, pp. 84–90, May 2017.

[44] S. Ioffe and C. Szegedy, "Batch normalization: Accelerating deep network training by reducing internal covariate shift," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2015, pp. 448–456.

[45] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," 2014, *arXiv:1409.1556*.

[46] A. G. Roy, N. Navab, and C. Wachinger, "Concurrent spatial and channel 'squeeze & excitation' in fully convolutional networks," in *Proc. 21st Int. Conf. Med. Image Comput. Comput. Assist. Intervent. (MICCAI)*, Granada, Spain. Cham, Switzerland: Springer, Sep. 2018, pp. 421–429.

[47] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2999–3007.

[48] Y. Feng, J. Jiang, H. Xu, and J. Zheng, "Change detection on remote sensing images using dual-branch multilevel intertemporal network," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4401015.

[49] Z. Li et al., "Lightweight remote sensing change detection with progressive feature aggregation and supervised attention," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5602812.

[50] A. Paszke et al., "PyTorch: An imperative style, high-performance deep learning library," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 32, 2019, pp. 8024–8035.

[51] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L.-C. Chen, "MobileNetV2: Inverted residuals and linear bottlenecks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 4510–4520.



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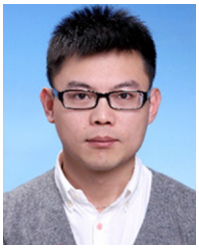
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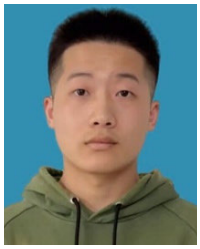
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